

DRAFT — DBA-VC Vendor Companion to the DBA Series

DBA-VC

Vendor Companion to the DBA Series

AI and Command Broker Requirements

Cross-Series Instrument

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Audience: ICS/OT Vendors, Suppliers, and System Integrators — Product Management, Engineering,
and Business Development

1. Purpose and Scope

This document is addressed to the vendors, suppliers, and system integrators who build and deliver industrial control systems (ICS) and operational technology (OT) products to critical infrastructure asset owners and operators. Its purpose is direct: to translate the AI and Command Broker requirements embedded in the DBA series into product specifications that a product manager, an engineering lead, or a business development executive can act on.

The DBA series — the Design Basis series published by Eclectic Technologies — extends the Design Basis Accident methodology of nuclear safety engineering across critical infrastructure sectors. Sector framework documents have been published for water (DBA-W), natural gas (DBA-GAS), oil (DBA-OIL), electricity (DBA-ES), data centers (DBA-DC), and nuclear (DBA-MA). Each sector framework carries requirements for the deployment of machine learning (ML) and generative AI (Gen AI) in OT control systems. Those requirements are stated in sector-specific operational vocabulary. This document abstracts them into canonical, sector-neutral product specifications and maps each specification to the sector frameworks in which it appears.

This document is a companion to the entire DBA series — not to any single sector. The Bright Line and Command Broker architecture established in Part I applies wherever a vendor's product operates, regardless of the sector label on the facility. A vendor supplying an ICS platform that is deployed across energy, water, nuclear, and transportation facilities carries the same cross-sector obligation at every installation. Part I establishes that obligation in sector-neutral terms. Part II instantiates it for the two most mature DBA sectors — energy and nuclear — and provides framework stubs for sectors where DBA work is in progress.

1.1 What This Document Is Not

This document does not establish a certification scheme. No DBA-series certification program currently exists. A vendor who builds to the specifications in Part I cannot yet receive a compliance certification, but can assemble the evidentiary package described in §6 that will support such a claim when procurement language begins to reference DBA-series requirements explicitly.

This document does not replace the sector framework documents. Vendors serving specific sectors should read the applicable sector framework for the full operational context of these requirements. DBA-VC provides the abstracted, cross-sector product specification; the sector frameworks provide the operational context in which the product will be deployed and assessed.

PART I — CROSS-SECTOR FOUNDATION

2. Analytical Foundation

The DBA series AI and Command Broker requirements derive from two analytical sources. The first is the risk-landscape study *Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers* (Roxey, T.E., Version 29, Eclectic Technologies, 2026). The second is the Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector v0.2), published by Eclectic Technologies, and its implementing instruments — the CB-IB Qualification and Onboarding Standard (v0.1), the Command Broker Implementation Guide (V0.2), and the Command Broker use-case series (CB-UC-1 v0.1). This section summarizes the key distinctions that govern product design requirements.

2.1 The ML / Gen AI Distinction

Machine learning (ML) encompasses algorithms that learn from data to improve performance — anomaly detection, predictive maintenance, pattern recognition, and load forecasting. ML algorithms can be deterministic: given bounded inputs, they produce bounded, testable outputs verifiable through semi-formal methods. An anomaly detection algorithm running on a SCADA historian is ML. It may operate autonomously without a Command Broker, because its behavior is bounded and testable.

Generative AI (Gen AI) encompasses large language models and similar systems that produce novel outputs from learned distributions. Gen AI is non-deterministic: the same input can produce different outputs across execution cycles, and formal verification methods do not apply. A large language model assistant proposed for HMI integration or operational decision support is Gen AI. It requires a Command Broker for any output that influences a physical process parameter at the highest-criticality band.

The vendor's obligation is to know which category their product occupies and to design accordingly. A product that combines ML and Gen AI components must assess each component independently against the band classification in §2.2.

2.2 The Criticality Bands

The CB-Framework establishes three criticality bands based on the severity of the worst credible consequence, aggregated across the six consequence categories by taking the maximum (Consequence-Severity Tier Model). Band assignment is determined by the consequence of the application, not the capability of the AI component.

Highest-Criticality band. Applications whose malfunction could produce Design Basis-level consequences: population-scale public health impact, loss of critical infrastructure function, or life-safety consequences. No autonomous Gen AI action is permissible in this band. A human Command Broker must be in the control loop for every change to a physical process parameter that a Gen AI component recommends or influences. ML that is deterministic and formally testable may operate autonomously in this band.

Mid-Criticality band. Applications involving moderate consequences that do not rise to the Design Basis threshold. Gen AI may inform operators; operators decide which action to execute. A human remains in the command loop. ML may operate autonomously.

Lowest-Criticality band. Applications whose consequences do not affect critical functions or life-safety. Gen AI may operate as a virtual assistant with limited autonomous action. ML operates with full autonomous behavior.

The band classification of an application is the joint responsibility of the vendor and the asset owner/operator (AOO). The vendor must document the band classification for which their product is designed and the configuration changes required to deploy at a different band. The AOO assigns the final band classification for their facility’s specific application context. The commissioning gate at which this joint classification is recorded and signed off is specified in CB-CT Commissioning & Joint Training (§4.1).

Naming crosswalk (legacy reading aid). The band names used here are canonical. Earlier documents in the corpus, and the ISA Parent Framework, numbered the same bands Tier 1 / 2 / 3; that numbering is retained below only to read older text. The parenthetical type-words — safety-critical, quality-of-service, administrative — are not band definitions; they name error classes and are defined as such in the Tiered Assessment Framework.

Criticality band (canonical)	Legacy number	Severity — worst credible consequence, any category
Highest	Tier 1	Design-Basis level: population- or mission-scale, loss of a critical function, or life-safety
Mid	Tier 2	Significant but bounded and recoverable, below the Design-Basis threshold
Lowest	Tier 3	No reach to a critical function, life-safety, or critical service in any category

2.3 The Bright Line

The Bright Line is a consequence threshold, not a technology classification or a fixed layer boundary. It is the line — drawn action by action — between actions whose erroneous execution could produce a Design Basis-level consequence, which must pass through a human Command Broker, and actions whose erroneous execution produces a correctable, facility-local consequence, which may execute under a lighter confirmation. Nothing above the line reaches physical command execution without passing through a human Command Broker, and that separation is enforced by architecture, not by procedure or operator discipline.

Two axes must be kept distinct, because a vendor that conflates them builds the wrong product. Band classification (§2.2) is an application-level judgment: it determines whether an application requires the Command Broker architecture at all. Bright Line placement is an action-level judgment made within a governed application: a single highest-criticality application routinely contains both above-line actions, routed through the broker, and below-line actions, executed under a lighter confirmation. Under an earlier reading a vendor built a single gate at the advisory/command seam; under this model the vendor builds a consequence classifier that routes action by action — the architecture developed in the Command Broker use-case series (CB-UC-1 §3) and instantiated for the EMS environment in §8.3. This application-versus-action distinction is separate from the criticality-band-versus-error-class distinction defined in the Tiered Assessment Framework; the two should not be merged.

The Bright Line is not a configuration option, a policy statement, or a vendor assurance. It is an engineering requirement: the control system must be mechanically incapable of executing a physical process command originating from a Gen AI advisory component without a positive human confirmation step. The confirmation step must be local, auditable, and not itself executable remotely or by another software component.

Note on naming: The term “Bright Line” is used exclusively in the AI/ML architectural boundary sense throughout all current DBA-series instruments. In the DBA-GAS and DBA-OIL sector frameworks, federal coordination trigger language uses “Federal Coordination Threshold,” not “Bright Line.” These are distinct concepts with distinct terms.

2.4 Cross-Sector ICS Amplifier

Many ICS and OT vendors do not serve a single sector. A SCADA platform, a distributed control system, an HMI suite, or a Gen AI operator decision support tool may be deployed across energy facilities, water treatment plants, nuclear power stations, transportation infrastructure, and data centers — often by the same vendor, sometimes in the same product line. This is the cross-sector amplifier: the Bright Line and Command Broker architecture follows the product, not the sector boundary.

The commercial implication is symmetrical. A vendor who builds to the DBA-series standard — Command Broker enforcement active by default, documented downgrade pathway, adversarial ML robustness specifications — has a compliance posture that applies across every sector in which their product is deployed. The qualification record built for a DBA-EN (energy) deployment is directly transferable to a DBA-W (water) or DBA-MA (nuclear) deployment. Vendors who build to the standard once do not rebuild sector by sector. Vendors who do not build to the standard carry a compliance deficit that compounds across every sector they serve.

The practical market signal: as DBA-series language propagates through engineering firm practice and utility procurement specifications, it will appear across sectors near-simultaneously. A vendor who has completed the evidentiary package in §6 is positioned to respond. A vendor

who has not must retrofit under procurement deadline pressure, simultaneously, across multiple sectors.

2.5 Vendor Applicability Self-Assessment

Not every vendor product carries the same DBA-series obligation. The obligation scales with the product’s proximity to physical command execution and the criticality of the application. This section provides narrative guidance for self-assessment, followed by a one-page checklist for quick triage.

The self-assessment proceeds in four steps. First, identify whether the product contains any AI component — ML or Gen AI — that has any pathway, direct or through integration, to a physical process parameter. If the product has no such pathway, the AI governance requirements in Part I do not apply to it, though supply chain assurance requirements may still apply to embedded software components.

Second, if a pathway exists, determine the criticality band of the application context. A product deployed in a water treatment disinfection system, a pipeline compressor station, a nuclear reactor protection system interface, or an energy management system dispatch function is in a highest-criticality application. The full Part I requirements apply.

Third, determine whether the AI component is ML or Gen AI. ML components that are deterministic and formally testable may operate autonomously below the Bright Line. Gen AI components require Command Broker interposition at the highest-criticality band. Products that combine both must assess each component independently.

Fourth, identify which sector-specific chapters in Part II apply to the product’s deployment contexts. A vendor serving only water utilities reads §10.1 and the DBA-W references. A vendor serving energy and nuclear reads §8 (Energy Sector) and §9 (Nuclear Sector). A vendor serving multiple sectors reads all applicable chapters and treats the most stringent requirement as the floor.

Vendor Applicability Checklist

Question	If Yes →
Does your product contain any AI component (ML or Gen AI) with any pathway to a physical process parameter?	Continue to the next question. If No: supply chain assurance requirements may still apply to embedded software.
Does your product operate in a highest-criticality application (water treatment, pipeline, nuclear, energy dispatch, or equivalent)?	Full Part I requirements apply. Proceed to §3 through §6.
Does your product operate above the Bright Line in any facility class covered by this series?	Command Broker enforcement is mandatory. See §3.1 and §3.2.

Does your product contain Gen AI components (non-deterministic, not formally testable)?	Command Broker required for the highest-criticality applications. Red-team obligation applies. See §3.1, §3.3, §3.6.
Does your product serve multiple sectors (energy, water, nuclear, transportation)?	Cross-sector amplifier applies. The most stringent sector requirement is the floor. See §2.4 and applicable Part II chapters.
Is your product an EMS platform serving Balancing Authority, RC, or TOP functions?	The Energy Sector chapter applies with CntlCo-specific depth. See Part II §8.
Does your product include I&C systems, safety system interfaces, or digital protection systems for nuclear facilities?	The Nuclear Sector chapter applies. See Part II §9.

3. Canonical Product Requirements

This section states each AI and Command Broker product requirement in canonical, sector-neutral form. Each requirement is organized into three obligation levels reflecting the default/downgrade/risk-transfer framework:

- **DEFAULT (Must):** What the product must do out of the box, without any configuration action by the AOO or integrator. This is the vendor’s design obligation.
- **DOWNGRADE PATHWAY (Should):** The conditions under which the default may be modified, and what the vendor must do to make that modification a documented, risk-visible act.
- **RISK TRANSFER OBLIGATION:** What the vendor’s product documentation must state when a downgrade is executed.

The Default/Should structure is not a weakening of the requirement. It is a recognition that the AOO holds authority and accountability for its facility’s design basis compliance. A vendor who ships with enforcement on by default, and who requires a positive documented configuration action to downgrade, has discharged their design obligation.

3.1 Command Broker Enforcement

DEFAULT (Must): The product ships with Command Broker enforcement active. The Gen AI component has no pathway to physical command execution without a positive human confirmation step that is local, auditable, and not executable remotely or by another software component.

DOWNGRADE PATHWAY (Should): The product may include a configuration option allowing the AOO or integrator to modify the enforcement level. Activation must require a

positive, deliberate configuration action — not a silent default, not a single-click override. The configuration action must generate a dated, auditable log entry.

RISK TRANSFER OBLIGATION: When a downgrade is executed, documentation must state: (a) that Command Broker enforcement has been reduced from the DBA-series default; (b) the specific residual risk — Gen AI-generated recommendations may reach physical command execution without independent human verification; (c) that the AOO assumes the compliance obligation; and (d) the AOO’s identity and date of authorization.

Conformance and misrepresentation. Command Broker conformance is assessed by the deploying organization, not asserted by the vendor; a vendor's compliance claim is not a substitute for the AOO's assessment (CB-UC-1 §6). A correctly implemented interface that routes recommendations through a human who rubber-stamps every output is architecturally compliant and functionally worthless. Marketing the presence of a confirmation dialog as, in itself, Command Broker compliance — where the surrounding workflow, time allocation, or authority structure does not permit genuine independent evaluation — is a material misrepresentation of DBA-series compliance, distinct from and additional to any downgrade disclosed under this section.

Which instrument governs. This document governs the product specification; the Command Broker Implementation Guide governs the asset owner’s obligations for qualifying and sustaining the personnel who perform the Command Broker function. A vendor conforms to DBA-VC, not to the Implementation Guide, and is not asked to satisfy both instruments independently. The vendor’s obligation with respect to the Implementation Guide is to supply the inputs the asset owner needs to discharge its own obligations — the vendor-supplied artifacts stated in §3.5.

3.2 Bright Line Architectural Separation

DEFAULT (Must): The product architecture enforces the Bright Line mechanically. The Gen AI advisory layer and the physical command layer are architecturally separated such that no Gen AI output reaches command execution without human interposition. This separation is verifiable through product documentation and independent testing.

DOWNGRADE PATHWAY (Should): Where an integrator or AOO requires architectural modification that reduces the separation, the vendor must provide written guidance stating that such integration violates the DBA-series Bright Line requirement for highest-criticality applications and that the compliance obligation transfers to the integrator or AOO.

RISK TRANSFER OBLIGATION: Vendor product datasheets, integration guides, and API documentation must not represent the Gen AI component as capable of “autonomous process optimization” or “direct control integration,” implying autonomous command authority at the highest-criticality band, without a prominent disclosure that such capability is inconsistent with

DBA-series compliance unless the Command Broker enforcement requirement in §3.1 is satisfied.

3.3 ML/Gen AI Component Assessment — The Three-Question Test

Before deploying any ML or Gen AI component in a highest-criticality application, the asset owner must be able to answer three questions about the product. The vendor's documentation obligation is to make the answers available without requiring the AOO to conduct an independent investigation.

Question 1: Does the ML or Gen AI component have any pathway to physical command execution without human confirmation? The vendor must document every pathway and confirm in writing whether each requires human confirmation.

Question 2: Is the component deterministic and formally testable? The vendor must classify the component as ML or Gen AI. Mixed products must classify each component independently.

Question 3: Has the Gen AI component been subjected to red-teaming? Red-teaming must be conducted by parties independent of the vendor. The vendor must document scope, methodology, and findings.

DEFAULT (Must): Product documentation provides clear, unambiguous answers to all three questions without requiring AOO investigation.

DOWNGRADE PATHWAY (Should): Where documentation does not yet address all three questions, the vendor should provide a compliance roadmap with dated commitments for each element.

RISK TRANSFER OBLIGATION: Vendor representations about product safety or compliance that are inconsistent with the three-question answers create warranty and liability exposure.

3.4 Operator-Facing Gen AI Workflow Constraints

DEFAULT (Must): The product does not pre-populate control system inputs from Gen AI output. The product does not present Gen AI recommendations as actionable defaults for the highest-criticality process parameters. The workflow requires an independent verification step between Gen AI advisory output and operator execution of a physical process command.

The product must also preserve the operator's independence from the advisory system itself. Independent system state monitoring must remain continuously visible on the primary operational display and must not be removed or obscured by the advisory interface at any time; the Gen AI display must never be the operator's only source of system state information. The operator must be able to dismiss, minimize, or override any recommendation without operational consequence — dismissal of a recommendation is not an audit event, confirmation of one is.

Visual and auditory alarm treatments for Gen AI recommendations must be configurable and must default to a lower urgency level than primary system alarms.

DOWNGRADE PATHWAY (Should): Where AOO workflow design reduces the independence of the verification step, the vendor's integration guidance must flag this as a Bright Line risk and recommend corrective workflow design.

RISK TRANSFER OBLIGATION: Cloud-connected Gen AI tools that have any pathway reaching the OT network create a network boundary risk independent of the Command Broker concern. Vendor documentation must disclose any network connectivity pathways that could reach OT systems.

3.5 Model Provenance and Supply Chain Assurance

DEFAULT (Must): The vendor maintains and provides to AOOs upon request: documented verification of training data sources; version control with change logging for model updates; vendor certification of update integrity; and a defined rollback process allowing the AOO to revert to a prior model version.

DEFAULT (Must), continued: The vendor also provides (e) an operator-facing AI system operational profile summary — process domain, training scope, and known limitations — in a form the Command Broker can read and apply during evaluation and qualification; (f) proactive advance notification to the AOO of any model update, fine-tuning change, or change to retrieval-augmentation sources, in a form and on a timeline that lets the AOO complete a broker requalification review before the changed system is placed in front of a broker; and (g) a qualification exercise mode that lets the AOO inject known-erroneous recommendations for broker qualification and continuing-competency exercises without affecting the production process.

DOWNGRADE PATHWAY (Should): Where the vendor's model update process does not yet include all four provenance elements, the vendor should provide a phased implementation roadmap, prioritizing integrity certification and rollback capability.

RISK TRANSFER OBLIGATION: Model updates delivered without vendor certification of update integrity place the AOO in the position of deploying an unverified change to a highest-criticality application. The vendor's update release documentation must state whether integrity certification is provided.

3.6 Red-Teaming Obligations

DEFAULT (Must): The vendor has subjected each Gen AI component intended for the highest-criticality ICS applications to red-team assessment by a party independent of the vendor, covering at minimum: prompt injection attacks; training data poisoning; adversarial inputs

designed to produce unsafe process parameter recommendations; and bypass of the Command Broker confirmation step.

DOWNGRADE PATHWAY (Should): Where independent red-teaming has not yet been completed, the vendor should clearly mark the product as pending red-team validation and provide a dated commitment to complete the assessment.

RISK TRANSFER OBLIGATION: Vendor representations that a Gen AI component is suitable for the highest-criticality ICS deployment without a documented independent red-team assessment create a material misrepresentation risk.

3.7 Audit Record Requirements

Every Command Broker confirmation, and every below-Bright-Line confirmation, produces an audit record. The record is the instrument by which a reviewer, a regulator, or an incident investigator later reconstructs why a broker acted, not merely that they did. This requirement abstracts CB-UC-1 §5 to sector-neutral form.

DEFAULT (Must): The product writes an audit record for every confirmation, in an open, non-proprietary, machine- and human-readable format with a documented schema (R-AUD-1); stores it in a system logically independent of the Gen AI advisory component and accessible to the AOO's compliance and legal staff independently of the vendor's tooling (R-AUD-2); makes it tamper-evident and non-modifiable after writing, with read access logged (R-AUD-4); and retains it for at least seven years or the applicable sector retention period, whichever is longer (R-AUD-3). Execution of a confirmed action is contingent on the successful write of its audit record (R-AL-4).

DOWNGRADE PATHWAY (Should): Where the product's audit store is not yet independent of the advisory component, or not yet in an open documented format, the vendor provides a phased roadmap prioritizing independent, AOO-accessible storage and an open export format.

RISK TRANSFER OBLIGATION: When audit records are retrievable only through the vendor's own platform or tooling, documentation must state: (a) that the independent-access requirement (R-AUD-2) is not satisfied; (b) that the AOO cannot independently reconstruct broker decisions without vendor cooperation; and (c) the AOO's acknowledgment and date of authorization. A vendor whose audit trail lives solely inside its own platform does not satisfy this requirement.

Traceability. Each canonical Part I requirement is governed by a section of the Command Broker corpus, as follows:

Canonical requirement (Part I)	Governing CB instrument section
§3.1 Command Broker Enforcement	CB-Framework §3.1–3.2; CB-IB Qualification Standard §2 (accountability principle)
§3.2 Bright Line Architectural Separation	CB-Framework §3.2; CB-UC-1 §4 (R-GEN, R-IND)
§3.3 Three-Question Test	CB-IB Qualification Standard §4 (competency domains)

§3.4 Operator-Facing Workflow Constraints	CB-UC-1 §4.5 (R-IND); §4.2–4.3
§3.5 Model Provenance & Supply Chain	Command Broker Implementation Guide §5.2, §8 (provenance documentation; update-triggered requalification)
§3.6 Red-Teaming	Command Broker Implementation Guide §7 (adversarial-realism exercises); CB-IB Qualification Standard §4.1 (adversarial scenario awareness)
§3.7 Audit Record Requirements	CB-UC-1 §5 (R-AUD-1...4) and §4.3 (R-AL-4); CB-IB Qualification Standard §7

4. Sector Applicability Matrix

The following table maps each canonical requirement in §3 to the sector frameworks where it appears. “Anticipated” indicates that the requirement is established as a series standard and will be incorporated into the named sector framework on its next substantive revision.

Requirement	DBA-W Water	DBA-GAS	DBA-OIL	DBA-ES Electric	DBA-MA Nuclear	EMS (CntlCo)
3.1 Command Broker Enforcement	§5.5.2	§5.4	§5.4	§3.4 (anticipated)	Safety system I&C boundary	§7.1.2 (licensed operator mandate)
3.2 Bright Line Architectural Separation	§5.5	§5.4	§5.4	§3.4 (anticipated)	Safety/non-safety network boundary	§7.1.1; §7.2
3.3 Three-Question Assessment	§5.5.3	§5.4	§5.4	Anticipated	Fourth question: software qualification	§7.1.3 (fourth question: adversarial injection)
3.4 Workflow Constraints	§5.5.4	Anticipated	Anticipated	Anticipated	Anticipated	§7.1.2 (role-based enforcement)
3.5 Model Provenance	§5.5.5	Anticipated	Anticipated	Anticipated	Software qualification pedigree	Anticipated
3.6 Red-Teaming	§5.5.3	§5.4	§5.4	Anticipated	IV&V requirement analog	§7.1.3 (adversarial injection testing)

5. The Procurement Signal

The AI Governance Series Overview §6.1 states the market reality directly: the pace of AI deployment in critical infrastructure is outrunning the standards process, but the standards process is moving. NERC is aware of the gap. NRC is aware of the gap. ISA and IEEE are actively developing standards. Vendors who build to the Bright Line before it becomes mandatory are building a qualification record and a product architecture that will survive the regulatory transition.

5.1 The Propagation Cascade

Step 1 — Series publication and engineering firm adoption. Engineering and consulting firms that serve critical infrastructure sectors adopt the DBA framework as their internal assessment standard. When a firm uses DBA-series event categories in a client assessment, it is applying the DBA standard without a regulatory mandate.

Step 2 — Utility procurement language. Asset owners whose assessments have been conducted against DBA-series standards begin incorporating DBA language into their procurement specifications. This happens across sectors near-simultaneously as the series matures.

Step 3 — Vendor product differentiation. Vendors who have built to the DBA-series specification can respond to procurement language immediately. Vendors who have not must retrofit — under procurement deadline pressure, across multiple sectors simultaneously.

Step 4 — Regulatory incorporation. The DBA series is designed for regulatory adoption. The water sector pathway runs through EPA’s AWIA resilience assessment guidance. The electric sector pathway runs through NERC reliability standards. The nuclear sector pathway runs through NRC rulemaking.

5.2 The Default/Downgrade Structure as a Market Signal

A vendor whose product ships with Command Broker enforcement on by default is telling the market: we built to the standard. A vendor whose product ships with enforcement off by default, or who offers enforcement as an add-on module, is telling the market: we built to a different standard. As procurement language matures, the default state question will become a standard due diligence item in vendor evaluation.

5.3 The Vendor Compliance Narrative

Vendors who build to the DBA-VC specification have a specific compliance narrative available:

- Our Gen AI components ship with Command Broker enforcement active by default. No configuration action by the AOO is required to achieve DBA-series alignment.
- We provide documented answers to the three-question assessment (§3.3) in our standard product documentation.

- We have conducted an independent red-team assessment of our Gen AI components against sector-specific attack scenarios.
- We maintain model provenance records and provide rollback capability for all Gen AI components in OT applications.
- When an AOO or integrator requests a downgrade, we provide documented risk transfer records that make the AOO's compliance obligation explicit.

This narrative is not marketing language — it is an evidentiary claim. Each element is verifiable by an AOO, an engineering firm, a regulator, or an incident investigator.

6. Path to Compliance Demonstration

No DBA-series certification program currently exists. What vendors can do is assemble an evidentiary package that supports a credible compliance claim now and positions the organization to transition to formal certification when a program is established.

6.1 The Evidentiary Package

Product Architecture Documentation. A document describing the Bright Line enforcement mechanism — how it is implemented, how it is verified, and what the default configuration state is. Sufficient detail to allow independent assessment without vendor-guided demonstration.

ML/Gen AI Component Classification. A classification document listing each AI component, its category (ML or Gen AI), its band classification, and its command authority pathway. Updated at each product release.

Three-Question Assessment Results. Vendor-generated answers to the three questions in §3.3, with supporting evidence.

Independent Red-Team Report. A report from a party independent of the vendor. Available to AOOs under NDA where commercially sensitive findings are included.

Model Provenance Records. Documentation of training data sources, version history, update integrity certification methodology, and rollback procedure.

Downgrade Configuration Procedure. A procedure document describing the steps required to execute a downgrade, including the configuration log entry generated and the risk transfer documentation produced.

AI System Operational Profile Summary. An operator-facing document stating the process domain, training scope, and known limitations of each Gen AI component, suitable for the Command Broker to apply during evaluation and qualification.

Model Update Notification Procedure. The procedure and timeline by which the vendor notifies the AOO of model updates, fine-tuning changes, and retrieval-source changes in advance of deployment.

Qualification Exercise Mode Description. Documentation of the product's mode for injecting known-erroneous recommendations to support broker qualification and continuing-competency exercises.

Commissioning sequence. These vendor-supplied artifacts — the operational profile summary, the model-update notification procedure, and the exercise mode — are placed in an ordered commissioning and re-qualification sequence, with joint training and gate sign-off, by the CB-CT Commissioning & Joint Training module (CB-CT_Commissioning_and_Joint_Training v0.2).

Audit Record Specification. The audit record schema, storage-independence design, tamper-evidence mechanism, retention period, and the AOO-independent access method (per §3.7).

6.2 Engagement Pathway

Vendors who wish to engage with the DBA series development process — to provide input on sector-specific product requirements, to participate in the development of FC instrument documents, or to be positioned as reference implementations for the Command Broker architecture — should contact Eclectic Technologies directly. The series is in active development. Vendor input on the translation from Design Basis requirement to product specification is valuable and welcome.

PART II — SECTOR CHAPTERS

7. Introduction to Sector Chapters

Part II instantiates the cross-sector foundation of Part I for each sector in which DBA instruments are active. Each chapter identifies the sector-specific facility classes, the highest-criticality application contexts, and the product requirements that go beyond the canonical requirements in §3 due to sector-specific operational or regulatory conditions. Vendors serving multiple sectors should read all applicable chapters and treat the most stringent requirement as the floor.

Part II currently contains substantive chapters for the two most analytically mature DBA sectors: Energy (§8) and Nuclear (§9). Framework stubs are provided for Water, Oil and Gas, and other sectors where DBA work is in progress. Those stubs will be expanded in subsequent revisions as the applicable DBA sector frameworks mature.

8. Energy Sector (DBA-EN)

8.1 Sector Overview

The DBA-EN Energy Sector series is the most extensively developed DBA sector series. It encompasses three sub-sectors — electric (DBA-ES), natural gas (DBA-GAS), and petroleum (DBA-OIL) — each with its own sector framework and facility-class instruments. The governing parent document is DBA-EN-SF (v0.8), which establishes the DB → DBA → DBA-MA three-layer progression as mandatory across all sub-sector series.

The energy sector presents the cross-sector amplifier argument with particular force. A vendor supplying an ICS platform for a pipeline compressor station may be selling to a facility that is simultaneously subject to DBA-GAS (natural gas sub-sector) and DBA-ES-GC-FC4 (the gas-electric cascade facility class, which propagates consequences through the electric sector). The Command Broker and Bright Line requirements apply at both the gas and electric consequence chain levels.

8.2 Highest-Criticality Applications — Energy Sector

Facility Class	Highest-Criticality Applications	Governing DBA Instrument
Balancing Authority / RC / TOP (CntlCo)	State estimator ML augmentation with AGC pathway; Gen AI operator decision support; bad data detection, adversarial robustness	DBA-ES-CntlCo v0.3 §7
EHV Transmission Substation (FC2)	Protection relay logic; automatic reclosing; transformer protection systems	DBA-ES-TX-FC2 v0.3
HVDC Interconnect (FC3)	Converter control systems; power flow management; automatic voltage regulation	DBA-ES-TX-FC3 v0.2
Gas Compressor Station (GC-FC4)	Compressor control and shutdown logic; pipeline pressure management; gas-electric cascade coordination	DBA-ES-GC-FC4 v0.4
Renewable Generation / BESS (GN-FC8)	Inverter control firmware; BESS charge/discharge management; grid-forming functions	DBA-ES-GN-FC8 v0.2
Natural Gas Transmission (GAS-FC1)	Compressor station operations; pipeline pressure management; SIS parameters; ESD logic	DBA-GAS-FC1 v1.6
Natural Gas Processing (GAS-FC2)	APC/RTO optimization-safety boundary; SIS trip point setbacks; NGL separation process parameters	DBA-GAS-FC2 v0.2
Petroleum Refining (OIL-	DCS process parameters; HF alkylation safety	DBA-OIL-FC1 v1.3

FC1)	controls; SIS functions	
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8.3 Energy Management System — CntlCo-Specific Requirements

This section translates the AI governance interaction points established in DBA-ES-CntlCo v0.2 §7 into product specifications for energy management system (EMS) vendors. These three interaction points are series firsts: they do not appear in any other DBA-ES instrument and carry design obligations specific to EMS products serving Balancing Authority (BA), Reliability Coordinator (RC), and Transmission Operator (TOP) functions.

8.3.1 State Estimator ML Augmentation — QA Obligations and the Gen AI/ML Layered Architecture

The Bright Line is a consequence threshold, not a technology classification. Actions whose erroneous execution could produce a BES reliability event are above the Bright Line; actions whose erroneous execution produces a correctable, facility-local consequence are below it. For state estimator ML augmentation in the EMS context, this distinction governs both what ML may do autonomously and where Gen AI fits into the architecture.

ML-augmented state estimators — bad data detection, topology processing, observability analysis — are deterministic and formally testable. They operate below the Bright Line and may execute autonomously in closed loop. The governing obligation is not advisory-only output but QA rigor: extensive testing, red-teaming against adversarial injection scenarios, and documented performance specifications are required before closed-loop autonomous operation is permitted. The ML system carries the determinism guarantee; the QA process validates that the guarantee holds at the envelope boundary.

Gen AI may participate in the state estimation and dispatch advisory chain, but only through a layered architecture: Gen AI passes a command to the ML implementation layer; the ML, operating within its QA-validated envelope, determines whether the command falls within its parameters and implements it if so. The Gen AI component never touches a physical process parameter directly. This layered architecture keeps the Gen AI below the Bright Line for actions whose consequences fall below the consequence threshold. Where a Gen AI-initiated command would produce an above-Bright-Line consequence — automatic redispatch, AGC setpoint modification, load shedding — the Command Broker intercepts before the command reaches the ML implementation layer.

The QA specification for ML components operating in closed-loop autonomous mode must separately address two distinct performance dimensions. The first is random error tolerance: bad data detection algorithm performance against random measurement errors, the standard specification that vendors currently provide. The second is adversarial injection tolerance: algorithm performance against coordinated false data injection at multiple measurement points

— the scenario in which an adversary crafts inputs to cause the ML system to produce a confident but false system state picture without triggering bad data detection. These are structurally different attack surfaces. A specification that addresses only random error tolerance does not satisfy the QA obligation for closed-loop autonomous ML operation in the CntlCo environment. This adversarial injection requirement is grounded in DBA-ES-CntlCo §7.1.3, which establishes that the MA-1 governing DBA-MA scenario — state estimator corruption via false data injection — is structurally identical to an adversarial ML attack on the state estimator’s inference inputs.

DEFAULT (Must): The EMS product ships with a documented QA package for each ML component operating in closed-loop autonomous mode. The package must separately specify: (a) random error tolerance — bad data detection algorithm performance against random measurement errors; and (b) adversarial injection tolerance — performance against coordinated false data injection at multiple measurement points. A single combined tolerance figure does not satisfy this requirement. Closed-loop autonomous ML operation is not permitted without both specifications on record. Gen AI components that pass commands to the ML implementation layer must have an architecturally enforced consequence classifier: commands that would produce above-Bright-Line consequences are routed to the Command Broker before reaching the ML layer; commands within the below-Bright-Line envelope may be passed to the ML layer for execution within its validated parameters.

DOWNGRADE PATHWAY (Should): Where an AOO deploys ML components without a complete QA package — including legacy state estimator deployments — the vendor should provide a compensating specification: the minimum testing and anomaly detection record that must be on file before closed-loop autonomous operation is authorized as an interim posture. A compensating package does not substitute for a full QA package; it is a documented interim authorization with a dated commitment to complete the full specification. Where the vendor’s adversarial injection tolerance assessment has not yet been completed, the vendor should provide a phased roadmap with a dated commitment.

RISK TRANSFER OBLIGATION: When a Gen AI component has a pathway to the ML implementation layer without a consequence classifier enforcing the above/below-Bright-Line routing, documentation must state: (a) that the consequence-classification gate required under this section is not in place; (b) that Gen AI-initiated commands may reach ML execution for above-Bright-Line consequences without Command Broker interposition; and (c) the AOO’s acknowledgment, the compensating controls in place, and the date of authorization. Vendor representations that an EMS’s bad data detection capability satisfies DBA-series requirements for CntlCo applications must be supported by both the random error tolerance and adversarial injection tolerance specifications. A representation based only on standard bad data detection performance does not constitute DBA-ES-CntlCo §7.1.3 compliance.

8.3.2 Gen AI EMS Operator Decision Support — Action-Specific Bright Line Placement and Licensed Operator as Command Broker

The Bright Line placement for a Gen AI operator decision support tool is action-specific, not tool-specific. The same EMS advisory system may present below-Bright-Line options — incremental economic dispatch within an existing operating range, battery dispatch within a defined envelope — and above-Bright-Line options — involuntary load shedding, actions affecting multiple BAs, irreversible system state changes — in a single operational view. The Command Broker requirement governs the above-Bright-Line options. It does not govern the tool's presence in the environment or its use for below-Bright-Line advisory functions.

For the CntlCo environment, DBA-ES-CntlCo establishes a specific Command Broker constraint: the Command Broker for any above-Bright-Line action must be the licensed system operator on duty. This follows from the NERC system operator certification standard and the operator's legal accountability under NERC's rules of procedure. The licensed operator's qualification domain — consequence assessment, system state evaluation, regulatory accountability — is what makes them the appropriate Command Broker for above-Bright-Line actions in this environment. Delegation to a non-licensed person is not permitted for above-Bright-Line confirmations.

DEFAULT (Must): The EMS Gen AI advisory product ships with a consequence classifier that routes each recommended action to either a below-Bright-Line confirmation pathway or an above-Bright-Line Command Broker confirmation pathway, and the interface is built to the full CB-UC-1 §4 requirement set. (a) Below-Bright-Line actions require a deliberate two-step select-then-confirm gesture (R-BL-2), with an audit record generated at confirmation (R-BL-4). (b) Above-Bright-Line actions require licensed system operator credential authentication and the full CB disclosure set in the option panel — consequence characterization, anomaly flags, confidence-and-basis summary, and the action boundary statement (R-AL-1); the action boundary statement is retained, not displaced by the credential requirement, because it is what keeps an authorization from being read as authorizing a class of similar actions rather than this action, at this magnitude, under this system state. (c) The above-Bright-Line confirmation enforces a deliberate review pause — a minimum display time of the disclosure fields, not less than five seconds and configurable upward by the AOO — before the confirmation input becomes available (R-AL-2), and the pause resets whenever a material change in conditions alters the consequence characterization or anomaly status of the pending option (R-MIX-3). (d) Options whose underlying system state has gone stale are visually flagged and their confirmation input disabled until the operator explicitly acknowledges the staleness (R-GEN-4, R-MIX-4). (e) An emergency override is provided: under an identified emergency the operator may suspend the review pause through a distinct, explicitly labeled override input that generates a distinct audit flag; the override compresses the confirmation but does not reduce the operator's Command Broker accountability (R-AL-3). A product built without

this pathway forces the wrong resolution to a fast-developing frequency excursion — moving actions below the line — which CB-UC-1 §3.3 explicitly declines to do. (f) Operator independence is preserved (R-IND-1 through R-IND-3), instantiating the cross-sector requirement in Part I §3.4: the advisory display does not obscure the independent system-state monitoring the operator needs; dismissing a recommendation is consequence-free and is not an audit event, while confirming one is; and Gen AI recommendations default to a lower urgency treatment than primary system alarms. (g) Execution of any confirmed action is contingent on the successful write of its audit record (R-AL-4).

DOWNGRADE PATHWAY (Should): Where an AOO configures the consequence classifier such that actions the DBA-ES-CntlCo instrument places above the Bright Line are routed to the below-Bright-Line confirmation pathway, the vendor must require a positive documented configuration action identifying the specific action categories reclassified, the analytical basis for the reclassification, and the compensating oversight architecture in place.

RISK TRANSFER OBLIGATION: When above-Bright-Line actions — as defined in DBA-ES-CntlCo — are confirmable without licensed operator credential authentication, documentation must state: (a) that the Command Broker qualification requirement under DBA-ES-CntlCo §7.1.2 is not satisfied for the affected action categories; (b) the specific actions and the reclassification basis; and (c) the AOO’s acknowledgment and date of authorization.

8.4 Inverter Firmware Trustworthiness — DBA-ES-GN-FC8

DBA-ES-GN-FC8 establishes inverter firmware trustworthiness as a Bright Line precondition: an inverter operating on unverified firmware cannot be treated as a below-Bright-Line autonomous ML system regardless of apparent normal behavior. Independent firmware verification is a precondition for autonomous inverter-based resource (IBR) operation. This applies to approximately 80–85% of deployed U.S. utility inverters by manufacturer origin.

For inverter vendors and IBR control system vendors, the product requirement follows directly: the vendor must provide a documented firmware verification pathway that allows the AOO or an independent assessor to confirm firmware integrity without relying solely on the vendor’s supply chain assurance. The three-question test in §3.3 applies to the inverter control firmware as an ML component: it is deterministic and formally testable only if the firmware has not been compromised. Trustworthiness — supply chain verified, not merely assumed — is the precondition for the determinism claim.

DEFAULT (Must): The inverter or IBR control system product ships with a documented firmware verification pathway. The pathway must allow the AOO or an independent third-party assessor to confirm firmware integrity through a mechanism independent of the vendor’s own supply chain assurance — including cryptographic hash verification against a vendor-maintained, externally auditable manifest. Firmware versions that cannot be independently

verified do not satisfy the Bright Line precondition for autonomous IBR operation under DBA-ES-GN-FC8.

DOWNGRADE PATHWAY (Should): Where an AOO operates inverters that lack an independent verification pathway — including legacy deployed units — the vendor should provide a compensating control specification: the minimum monitoring and anomaly detection architecture that must be in place before the inverter may be treated as below-Bright-Line for operational purposes. A compensating control package is not equivalent to verified firmware; it is a documented interim posture pending verification pathway availability.

RISK TRANSFER OBLIGATION: When an inverter or IBR control product is deployed without a completed independent firmware verification, documentation must state: (a) that the firmware trustworthiness precondition under DBA-ES-GN-FC8 has not been satisfied; (b) that the inverter must be treated as above-Bright-Line pending verification, requiring Command Broker governance for any autonomous control actions; and (c) the AOO's acknowledgment and date of authorization.

8.5 Natural Gas Processing — Optimization-Safety Boundary

DBA-GAS-FC2 establishes the optimization-safety boundary as a formal Design Basis requirement — a series first. In continuous-process gas processing facilities, the Bright Line must constrain the plant's own commercial optimization logic, not only adversarial manipulation. Advanced Process Control (APC) and Real-Time Optimization (RTO) envelopes must include safety margin setbacks from SIS trip points, enforced by architecture.

For APC, RTO, and process optimization vendors serving natural gas processing facilities, this creates a product requirement that does not exist in traditional OT cybersecurity frameworks: the commercial optimization algorithm itself is a Bright Line candidate. If APC/RTO output can modify process parameters into the safety margin band without a safety instrumented system interlock or operator confirmation gate, the optimization system is operating above the Bright Line. The vendor must document where the APC/RTO optimization envelope ends and the SIS trip point begins, and must confirm that the product architecture enforces the setback.

DEFAULT (Must): The APC/RTO product ships with a documented optimization envelope boundary specification that identifies, for each controlled process variable, the maximum setpoint the optimization algorithm may command without a safety interlock or operator confirmation gate. The boundary must be expressed as a measurable setback from the SIS trip point — not as a general design principle. The optimization envelope boundary specification is a mandatory deliverable in the product's engineering documentation package.

DEFAULT (Must), continued: The product must additionally disclose the optimization objective function and its relationship to the safety margin, so that the person authorizing movement along the optimization gradient can see the objective the system is driving toward. This is the control against gradual margin erosion — the case in which each

individual setpoint change is within bounds and correct on its own merits, but the aggregate trajectory walks the process toward the safety margin. Trend visibility across authorizations, not per-action review, is what catches it.

DOWNGRADE PATHWAY (Should): Where an AOO configures APC/RTO setpoints that reduce or eliminate the safety margin setback — for example, to maximize NGL recovery or throughput during peak demand — the vendor must require a positive, documented configuration action. The configuration record must identify the affected process variables, the modified setback margins, and the compensating SIS or operator confirmation architecture that replaces the default envelope boundary.

RISK TRANSFER OBLIGATION: When APC/RTO product output can reach SIS trip point proximity without a confirmed interlock or operator confirmation gate, documentation must state: (a) that the optimization-safety boundary requirement under DBA-GAS-FC2 is not satisfied by the default configuration; (b) the specific process variables and setback margins that have been modified; and (c) the AOO's acknowledgment, the compensating control architecture in place, and the date of authorization.

9. Nuclear Sector (DBA-MA)

9.1 Sector Overview

The DBA-MA Nuclear sector series is the analytical origin of the Design Basis methodology that the entire DBA series extends to energy and other critical infrastructure sectors. Nuclear safety engineering formalized the DB → DBA progression in the 1960s and 1970s. The DBA-EN series inherits that methodology and extends it to deliberate military action. The nuclear sector's governance framework is therefore the most mature in the DBA series, and its product requirements carry the deepest regulatory grounding.

Nuclear facility I&C vendors operate in a regulatory environment defined by 10 CFR 50 Appendix B (quality assurance for nuclear power plant safety-related systems), IEEE 603 (criteria for safety systems in nuclear power generating stations), IEEE 7-4.3.2 (application of digital computers to nuclear power plant safety systems), and the NRC's cybersecurity rule under 10 CFR 73.54. The DBA-MA nuclear sector requirements do not replace these regulatory requirements — they extend them to address military action threat scenarios that existing regulatory frameworks did not anticipate.

The Bright Line in the nuclear context has a direct analog in the existing regulatory framework: the safety/non-safety network boundary. IEEE 603 and 10 CFR 50 Appendix A General Design Criterion 24 establishes the principle that safety systems must be isolated from non-safety systems. The DBA-MA extension applies the same principle to AI and Gen AI components: a Gen AI advisory component is by definition a non-safety system and must not have a pathway to

safety system parameters without human interposition by a qualified operator. The Bright Line is the AI-era restatement of the isolation principle.

9.2 I&C Vendor Landscape

Nuclear I&C vendors fall into two primary categories for DBA-MA purposes. The first is the safety system I&C supplier — vendors of reactor protection systems, engineered safety feature actuation systems, and safety-related control systems. These vendors operate under 10 CFR 50 Appendix B quality assurance requirements and must demonstrate software qualification under IEEE 7-4.3.2. The second is the cybersecurity overlay vendor — vendors of OT security monitoring, network segmentation, anomaly detection, and incident response tools deployed in the nuclear OT environment. These vendors must operate within the 10 CFR 73.54 cybersecurity rule framework.

Gen AI tools are emerging in both categories: AI-assisted safety system diagnostics, Gen AI operator advisory tools for emergency procedure guidance, and ML-based anomaly detection for nuclear OT networks. The DBA-MA nuclear sector requirements address each category. The Bright Line applies with particular rigor in the nuclear context because the regulatory framework has already established the isolation principle — the DBA-MA extension makes the AI governance implication explicit.

9.3 Safety/Non-Safety Network Boundary as the Nuclear Bright Line

The nuclear analog to the Bright Line is the safety/non-safety network boundary established by IEEE 603 §5.1 and 10 CFR 50 Appendix A GDC-24. The principle is architectural: safety systems must be isolated from non-safety systems such that a failure in a non-safety system cannot prevent safety system action or cause spurious safety system actuation.

For AI and Gen AI components, this means:

- A Gen AI advisory component operating on the non-safety (balance of plant) network must have no electronic pathway to safety system parameters without human interposition by a licensed senior reactor operator (SRO). The SRO is the nuclear sector's Command Broker.
- An ML component operating within a safety system must be qualified under IEEE 7-4.3.2 as a software component of a safety-related system. The determinism and formal testability requirements of the Bright Line framework align with, and are subsumed by, the software qualification requirements of the nuclear regulatory framework.
- A cybersecurity monitoring tool operating on the non-safety network that generates alerts or recommendations related to safety system status must not have a direct pathway to safety system actuation or parameter change. The alert's downstream action — if any — must pass through the licensed operator.

DEFAULT (Must): I&C products intended for nuclear applications enforce the safety/non-safety network isolation as an architectural property. No Gen AI component has a pathway, direct or through integration, to a safety system parameter without SRO interposition.

DOWNGRADE PATHWAY (Should): Where an integrator or AOO proposes a deployment configuration that creates a pathway from a Gen AI advisory component to a safety system parameter without SRO interposition, the vendor must provide written guidance stating that such a configuration violates the nuclear Bright Line requirement and that any regulatory deviation must be evaluated under the 10 CFR 50.59 change control process.

RISK TRANSFER OBLIGATION: Vendor representations that a Gen AI or AI-assisted tool is suitable for safety-related nuclear applications must be supported by a software qualification pedigree under IEEE 7-4.3.2 or an explicit statement that the tool is classified as a non-safety system and is subject to the isolation requirement above.

9.4 Safety System Software Qualification

Safety-related software in nuclear applications must be qualified under IEEE 7-4.3.2. The qualification process includes: (1) software development lifecycle documentation; (2) formal verification and validation (V&V), including software hazard analysis; (3) independence requirements for V&V activities; and (4) configuration management throughout the software lifecycle. These requirements apply to any ML component integrated into a safety-related system.

The three-question test in §3.3 applies with a nuclear-specific fourth question for safety-related applications: Has the ML component been qualified under IEEE 7-4.3.2 or the applicable NRC-accepted alternative? The answer must be yes for any ML component embedded in a safety-related system. A Gen AI component — non-deterministic by definition — cannot be qualified under the existing nuclear software qualification framework and therefore cannot be integrated into a safety-related system under current regulatory requirements.

DEFAULT (Must): Vendors of safety-related I&C systems document the software qualification pedigree of all embedded ML components, including the applicable IEEE standard under which qualification was performed, the independent V&V organization, and the configuration management record.

DEFAULT (Must): Vendors of safety-related I&C systems do not integrate Gen AI components into safety-related software. Where a product includes both safety-related and non-safety functions, the Gen AI components are architecturally isolated from the safety-related software in a manner verifiable under 10 CFR 50 Appendix B.

DOWNGRADE PATHWAY: No downgrade pathway exists for the Gen AI exclusion from safety-related systems under the current regulatory framework. Any proposed integration of non-deterministic AI into a safety-related system requires an NRC exemption or rulemaking.

9.5 Cybersecurity Overlay Vendors — Nuclear OT

Vendors of cybersecurity tools deployed in the nuclear OT environment operate under the 10 CFR 73.54 cybersecurity rule. AI-based anomaly detection tools deployed in the nuclear OT environment carry the ML/Gen AI classification obligation from §2.1. An ML-based anomaly detection tool that is deterministic and formally testable may operate autonomously below the Bright Line. However, any response action triggered by the tool that could affect safety system function, security system function, or emergency preparedness system function requires operator interposition before execution.

DEFAULT (Must): Nuclear OT security monitoring tools ship with automated response actions that affect safety, security, or emergency preparedness functions disabled by default. Automated response actions in those consequence categories are available only as a positive AOO configuration choice, with documented authorization, auditable log entry, and risk transfer documentation consistent with §3.1.

DOWNGRADE PATHWAY (Should): Where an AOO enables automated response actions affecting safety-adjacent systems, the vendor must require a documented configuration action reviewed by the AOO's security organization and consistent with the facility's 10 CFR 73.54 cybersecurity plan.

9.6 Red-Teaming in the Nuclear Context

Red-teaming of Gen AI components in nuclear non-safety applications follows the requirements in §3.6, with the addition of a nuclear-specific attack scenario: the adversary's goal in a DBA-MA nuclear scenario is not simply to compromise a system but to cause a safety function to fail or to trigger spurious safety actuation while evading detection. Gen AI tools deployed in the nuclear environment must be red-teamed against scenarios in which the adversary has crafted inputs designed to produce recommendations that would lead an operator to take an action inconsistent with safety function requirements.

This is the nuclear sector analog to the adversarial injection requirement in §8.3.1: the adversary is not attacking the system directly but is attacking the information presented to the human Command Broker — the licensed SRO — to induce an unsafe decision. Red-team scope for nuclear Gen AI advisory tools must explicitly include this attack vector.

10. Emerging Sectors

The following sectors have active DBA development work in progress. Substantive vendor requirements for these sectors will be incorporated into future DBA-VC revisions as the applicable sector frameworks mature. Vendors serving these sectors should consult the applicable DBA framework documents directly and engage with Eclectic Technologies for current guidance.

10.1 Water Sector (DBA-W)

The DBA-W Sector Framework is the most mature non-energy, non-nuclear DBA instrument. §5.5 of DBA-W carries the most fully developed sector-specific implementation of the Command Broker and Bright Line requirements currently in the series. Vendors serving water sector clients should read DBA-W §5.5 as the current authoritative source for water-sector AI governance product requirements. A substantive water-sector chapter will be incorporated into DBA-VC on the next scheduled revision.

10.2 Oil and Gas Sector

The DBA-GAS and DBA-OIL sector frameworks carry Command Broker and Bright Line requirements in gas and petroleum sector operational vocabulary (§5.4 of each). The optimization-safety boundary requirement established in DBA-GAS-FC2 (see §8.5 of this document) is the most distinctive energy-sector-specific requirement for APC and RTO vendors. The sector applicability matrix in §4 provides the current reference mapping. A dedicated oil and gas sector chapter will be incorporated in a future revision.

10.3 Transportation and Other Critical Infrastructure Sectors

DBA series development for transportation, communications, and other NIPP critical infrastructure sectors is in the planning stage. Vendors serving these sectors who wish to engage with the DBA series development process should contact Eclectic Technologies directly.

References

DBA Series

Roxey, Timothy E. Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector v0.2). Eclectic Technologies, 2026.

Roxey, Timothy E. CB-IB Qualification and Onboarding Standard (CB-IB_Qualification_Standard v0.1). Eclectic Technologies, 2026.

Roxey, Timothy E. Command Broker Implementation Guide (V0.2). Eclectic Technologies, 2026.

Roxey, Timothy E. CB-UC-1: Command Broker Use Case 1 — Balancing Authority Integrated Interface (v0.1). Eclectic Technologies, 2026.

Roxey, Timothy E. CB-CT: Command Broker Commissioning & Joint Training (CB-CT_Commissioning_and_Joint_Training v0.2). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-EN-SF: Energy Sector Design Basis Framework (v0.8). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-ES-CntlCo: Electric Sector Control-Company Functional Class (v0.3). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-ES-GN-FC8: Renewable Generation and BESS (v0.2). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-ES-MA-SF: Electric Sector Military Action Sector Framework (v0.2). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-GAS-FC2: Natural Gas Processing (v0.2). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-MA Introduction (v1.9). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-W Sector Framework (v1.5). Eclectic Technologies, 2026.

Roxey, Timothy E. DBA-DC-L1 Design Basis Framework (v1.7). Eclectic Technologies, 2026.

Foundational Study

Roxey, Timothy E. Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers. Version 29. Eclectic Technologies, 2026. (Versions 21 and later reflect the author's continued independent development following the ISAGCA Supplier Working Group submission at Version 20.)

Standards

International Society of Automation / International Electrotechnical Commission. ISA/IEC 62443 Series: Security for Industrial Automation and Control Systems. Research Triangle Park: ISA (current editions).

IEEE Standard 603-2018: Criteria for Safety Systems for Nuclear Power Generating Stations.

IEEE Standard 7-4.3.2-2016: Application of Digital Computers to Nuclear Power Plant Safety Systems.

U.S. Nuclear Regulatory Commission. 10 CFR Part 73.54: Protection of Digital Computer and Communication Systems and Networks.

U.S. Nuclear Regulatory Commission. 10 CFR Part 50, Appendix B: Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants.

Regulatory

North American Electric Reliability Corporation. NERC CIP Reliability Standards (current versions).

U.S. Nuclear Regulatory Commission. Regulatory Guide 1.152 (Rev. 3): Criteria for Use of Computers in Safety Systems of Nuclear Power Plants.

Revision History

Ver.	Date	Author	Description
0.1	Jun 2026	T. Roxey	Initial working draft. Established cross-series vendor companion document type. Six canonical product requirements derived from DBA-W §5.5, DBA-GAS §5.4, and DBA-DC §3.5. Default/downgrade/risk-transfer obligation structure introduced. Sector applicability matrix and highest-criticality tier table completed. The procurement signal cascade argument has been developed. Evidentiary package elements identified.
0.2	Jun 2026	T. Roxey	Added §3.7: EMS CntCo-specific requirements. Three interaction points from DBA-ES-CntCo v0.2 §7 translated into product specifications: state estimator ML advisory-only boundary; Gen AI operator decision support with licensed operator as mandatory Command Broker; adversarial input robustness as a distinct product security requirement. Sector applicability matrix updated. §2.3 Bright Line note revised to reflect correction of “Federal Coordination Threshold” usage in DBA-GAS and DBA-OIL.
0.3	Jun 2026	T. Roxey	Major restructuring. Reorganized into Part I (Cross-Sector Foundation) and Part II (Sector Chapters). §2.4 Cross-Sector ICS Amplifier added to Part I as a core framing argument. §2.5 Vendor Applicability Self-Assessment added with narrative guidance and one-page checklist. Part II §8 (Energy Sector) established, absorbing prior §3.7 CntCo content into §8.3 and adding §8.4 (Inverter Firmware Trustworthiness) and §8.5 (Optimization-Safety Boundary). Part II §9 (Nuclear Sector) added as substantive chapter. Part II §10 (Emerging Sectors) added as framework stubs. Sector applicability matrix updated. References expanded to include the nuclear regulatory framework.
0.4	Jun 2026	T. Roxey	Energy chapter expanded. §8.4 (Inverter Firmware Trustworthiness) and §8.5 (Natural Gas Processing — Optimization-Safety Boundary) brought to full product specification standard: Default/Downgrade/Risk Transfer structure added to each. §8.1 and §8.2 version references updated (DBA-EN-SF v0.7 → v0.8; DBA-ES-TX-FC2 v0.2 → v0.3; DBA-ES-GN-FC8 v0.1 → v0.2). References updated to match. Gap N-19 closed (CntCo EMS vendor specification confirmed complete in v0.3 §8.3). Superseded by v0.5 due to errors in §8.3.1 and §8.3.2; deleted.
0.5	Jun 2026	T. Roxey	§8.3.1 and §8.3.2 revised to correctly reflect V29 §5.3 and CB-Framework Bright Line architecture. §8.3.1: Removed incorrect advisory-only restriction on ML. Established QA obligations (random error tolerance + adversarial injection tolerance) as the governing condition for closed-loop autonomous ML operation. Introduced Gen AI/ML layered architecture and consequence classifier as the Bright Line enforcement mechanism. Absorbed adversarial injection tolerance requirement (formerly §8.3.3) into §8.3.1 DEFAULT block. §8.3.2: Removed categorical “Gen AI tools operate above the Bright Line” statement. Established action-specific Bright Line placement per CB-UC-1. Rewrote DEFAULT block around consequence classifier and two-pathway confirmation architecture. Retained licensed operator as mandatory Command Broker for above-Bright-Line confirmations. §8.3.3 removed (content absorbed into §8.3.1).
0.6	Jul	T.	Absorbed CB-UC-1, the CB-IB Qualification Standard, and the

	2026	Roxey	Command Broker Implementation Guide as sources. §2.3 reconciled to the consequence-threshold model with the application-level (tier) vs. action-level (Bright Line placement) distinction stated explicitly; §2.2 tier crosswalk added. New canonical §3.7 (audit record requirements, from CB-UC-1 §5). §3.1 conformance-and-misrepresentation clause added. §3.5 extended with three vendor-supplied artifacts (operational profile summary, proactive update notification, qualification exercise mode); §6.1 evidentiary package updated to match. §8.3.2 DEFAULT rebuilt against the full CB-UC-1 §4 R-AL/R-MIX/R-IND set (action boundary statement restored; review pause, staleness flagging, emergency override, operator independence, and audit-write contingency added). §8.5 optimization-objective disclosure added. Source study citation corrected to the canonical Eclectic Technologies edition. Mechanical fixes: §9.6 reference §8.3.3 → §8.3.1; §2.5 water-sector reference §3.1 → §10.1; 'Fourth Edition' removed from cover. Traceability table added.
0.7	Jul 2026	T. Roxey	Reconciled to the Consequence-Severity Tier Model. §2.2 recast as criticality bands on the six-category take-the-max basis; naming crosswalk demoted to legacy and the type-words returned to the error classes; §2.3 band/action distinction clarified against the error-class axis; obligation structure relabeled 'obligation levels' to free 'tier' for criticality; band vocabulary applied throughout.
0.8	Jul 2026	T. Roxey	Restored the cross-sector operator-independence requirement to Part I §3.4 (recovered from the June v0.6 draft; §8.3.2 now references it as the energy instantiation). Restated the governs-which-instrument clause in §3.1 (a vendor conforms to DBA-VC, not the Implementation Guide; inputs supplied per §3.5) — the §2.3.1 Convergent Origination framing stays retired. Ran the version-citation pass: CB-Framework → v0.2, Implementation Guide → V0.2, CB-CT → v0.2.